accumulated human experience in chess, as well as to decades of computer experience. It was able to calculate 70 million chess positions per second. In contrast, AlphaZero performed only 80,000 such calculations per second, and its human creators never taught it any chess strategies – not even standard openings. Rather, AlphaZero used the latest machine-learning principles to self-learn chess by playing against itself. Nevertheless, out of a hundred games the novice AlphaZero played against Stockfish, AlphaZero won twenty-eight and tied seventy-two. It didn't lose even once. Since AlphaZero learned nothing from any human, many of its winning moves and strategies seemed unconventional to human eyes. They may well be considered creative, if not downright genius.

Can you guess how long it took AlphaZero to learn chess from scratch, prepare for the match against Stockfish, and develop its genius instincts? Four hours. That's not a typo. For centuries, chess was considered one of the crowning glories of human intelligence. AlphaZero went from utter ignorance to creative mastery in four hours, without the help of any human guide.[18]

AlphaZero is not the only imaginative software out there. Many programs now routinely outperform human chess players not just in brute calculation, but even in 'creativity'. In human-only chess tournaments, judges are constantly on the lookout for players who try to cheat by secretly getting help from computers. One of the ways to catch cheats is to monitor the level of originality players display. If they play an exceptionally creative move, the judges will often suspect that this cannot possibly be a human move – it must be a computer move. At least in chess, creativity is already the trademark of computers rather than humans! Hence if chess is our coal-mine canary, we are duly warned that the canary is dying. What is happening today to human–AI chess teams might happen down the road to human–AI teams in policing, medicine and banking too.[19]

Consequently, creating new jobs and retraining people to fill them will not be a one-off effort. The AI revolution won't be a single watershed event after which the job market will just settle into a new equilibrium. Rather, it will be a cascade of ever-bigger disruptions. Already today few employees expect to work in the same job for their entire life.[20] By 2050, not just the idea of 'a job for life', but even the idea of 'a profession for life' might seem antediluvian.

Even if we could constantly invent new jobs and retrain the workforce, we may wonder whether the average human will have the emotional stamina necessary for a life of such endless upheavals. Change is always stressful, and the hectic world of the early twenty-first century has produced a global epidemic of stress.[21] As the volatility of the job market and of individual careers increases, would people be able to cope? We would probably need far more effective stress-reduction techniques – ranging from drugs through neuro-feedback to meditation – to prevent the Sapiens mind from snapping.